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Factor Investing: The Value Premium

Replicating Fama & French (1992)

Fama, E. F. & French, K. R. (1992). The Cross-Section of Expected Stock Returns. Journal of Finance, 47(2), 427–465.

Topics covered in this lecture:

- Why cheap stocks (high BE/ME) earn higher returns — the value premium
- Constructing the Book-to-Market ratio without look-ahead bias
- Fama-MacBeth cross-sectional regressions — the methodology workhorse
- BE/ME and size decile portfolios: construction and backtesting
- Honest assessment: why value failed in 2010–2025 and what that means

Lecture Roadmap

1	**The Value Premium** *Economic intuition and two competing interpretations*	~15 min
2	**Data & Signal Construction** *BE/ME ratio, look-ahead-free merging, data filters*	~15 min
3	**Fama-MacBeth Regressions** *Cross-sectional methodology and Table IV replication*	~15 min
4	**Portfolio Construction** *Decile sorts and the VMG long-short strategy*	~10 min
5	**Backtesting** *Cumulative returns, performance by decile, sub-periods*	~15 min
6	**Paper vs Replication** *Side-by-side comparison of key exhibits*	~10 min
7	**Robustness Checks** *Sub-period, quintile, sector-neutral versions*	~5 min
8	**Honest Assessment** *What replicates, what doesn't, and why*	~5 min

1 The Value Premium

Book-to-market intuition and the two competing explanations

The Book-to-Market Ratio: Value vs. Growth

The Fundamental Question

- Why do stocks trading below their book value (cheap / high BE/ME) earn higher average returns than stocks trading at a premium to book (expensive / low BE/ME)?

Value Stock (High BE/ME)

- Company A: market price = $100, book value = $120 → BE/ME = 1.2.
- The market values it below what it owns — it looks cheap, unloved, or risky.
- Historically earns HIGHER average returns per Fama & French (1992).

Growth Stock (Low BE/ME)

- Company B: market price = $100, book value = $20 → BE/ME = 0.2.
- Market prices it far above book — investors pay a premium for expected future growth.
- Historically earns LOWER average returns per Fama & French (1992).

Fama & French (1992) Central Hypothesis

- BE/ME and size (market cap) JOINTLY capture the cross-section of U.S. stock returns — CAPM beta is irrelevant once these two variables are included.

Two Competing Interpretations of the Value Premium

Risk-Based Interpretation (FF's View)

- Value stocks are cheap because they are fundamentally RISKIER — more exposed to economic downturns, financial distress, and earnings uncertainty.
- Their higher returns are compensation for bearing systematic risk not captured by market beta.
- Implication: value stocks will suffer when you can least afford it (recessions, credit crises).

Behavioral Interpretation

- Investors SYSTEMATICALLY extrapolate past growth too far into the future, causing growth stocks to be overpriced and value stocks underpriced.
- The premium reflects mean reversion of mispricing, not compensation for risk.
- Implication: the premium is an arbitrage opportunity that sophisticated investors can exploit — and eventually will arbitrage away.

Key Critiques of the Original Finding

- Data mining: Lo & MacKinlay (1990) warned that searching many variables inflates false-discovery rates — value was not the only variable tested.
- Survivorship bias: early COMPUSTAT coverage skews toward survivors, overstating historical returns.
- Post-publication decay: value premium has weakened significantly post-1990, especially during 2008–2020.
- Intangibles problem: modern firms hold large off-balance-sheet assets (software, IP, brands) — BE/ME is a noisier signal today.

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Data & Signal Construction

BE/ME ratio, look-ahead-free merging, and universe filters

Data Description and Sample Characteristics

865	705	192	~685
Stock tickers (raw)	Non-financial stocks	Months (Jan 2010–Dec 2025)	Stocks with valid BE/ME

Three Data Sources

- us_stock_prices.parquet — daily adjusted close, 4,024 days × 865 stocks.
- company_financials.parquet — 66,149 rows, 1,328 stocks, with announce_date for each filing.
- stocks_basic_info.parquet — sector, shares outstanding (static snapshot).

Key Sample Filters

- Exclude financial-sector stocks (222 tickers) — matches FF's SIC 6000–6999 exclusion.
- Require BE > 0: negative book equity is economically uninterpretable.
- Winsorize BE/ME at 1st and 99th percentile each month — removes data errors.
- BE/ME mean = 0.451, median = 0.333, IQR: 0.083 – 0.903.

Shares outstanding treated as static. This introduces ME estimation noise but no systematic directional bias in cross-sectional rankings.

Constructing BE/ME Without Look-Ahead Bias

Monthly Return (log)

$$r_{\{i,t\}} = \ln(P_{\{i,t\}} / P_{\{i,t-1\}})$$

Market Capitalization

$$ME_{\{i,t\}} = P_{\{i,t\}} \times SharesOutstanding_i$$

Book-to-Market Ratio (no look-ahead)

$$BE/ME_{\{i,t\}} = CommonEquity_{\{i,t*\}} / ME_{\{i,t\}}$$

`where t* = latest announce_date ≤ t  (never use data announced after month t)`

Look-Ahead-Free Implementation

- Use announce_date (public filing date), NOT fiscal period end date.
- For each month-end t: take the most recently announced common equity filed on or before t.
- Forward-fill within the monthly grid — preserves the as-of merge logic.

Signal Timing

- BE/ME known at end of month t → predicts returns in month t+1.
- One-month lag applied before all portfolio sorts and regressions.
- This avoids execution bias: signal at t+1 open determines positions entering t+2.

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Fama-MacBeth Regressions

Two-step cross-sectional methodology and Table IV replication

Fama-MacBeth (1973) Regression Methodology

Monthly Cross-Sectional Regression (Step 1)

$$r_{\{i,t+1\}} = \alpha_t + \beta_t \cdot \ln(BE/ME)_{\{i,t\}} + \gamma_t \cdot \ln(ME)_{\{i,t\}} + \varepsilon_{\{i,t+1\}}$$

$$Run\ OLS\ cross-section\ each\ month\ t. Store\ \hat{\beta}_t, \hat{\gamma}_t.$$

Time-Series Summary (Step 2)

$Estimate = mean(\hat{\beta}_t)$ `over all months T. SE = Newey-West (6 lags) for autocorrelation.`
$t-stat = mean\ /\ NW-SE$. `This correctly handles cross-sectional correlation in returns.`

Why Not Pooled OLS?

- Returns in the same period are correlated across stocks (market moves everything together) — standard OLS t-stats are inflated.
- FM separates the problem: each month's cross-section is separate, and the final inference treats the monthly slopes as IID draws.
- Newey-West SE further corrects for autocorrelation in the coefficient series.

Paper Findings (Table IV)

- Sample: NYSE/AMEX/NASDAQ, July 1963 – Dec 1990 (342 months).
- log(BE/ME): coef = +0.50%/month, t-stat = 5.71 → highly significant, positive.
- log(ME): coef = −0.15%/month, t-stat = −2.58 → small caps earn more.
- Market beta: NOT significant once BE/ME and size are included.

Fama-MacBeth Results: Table IV Comparison

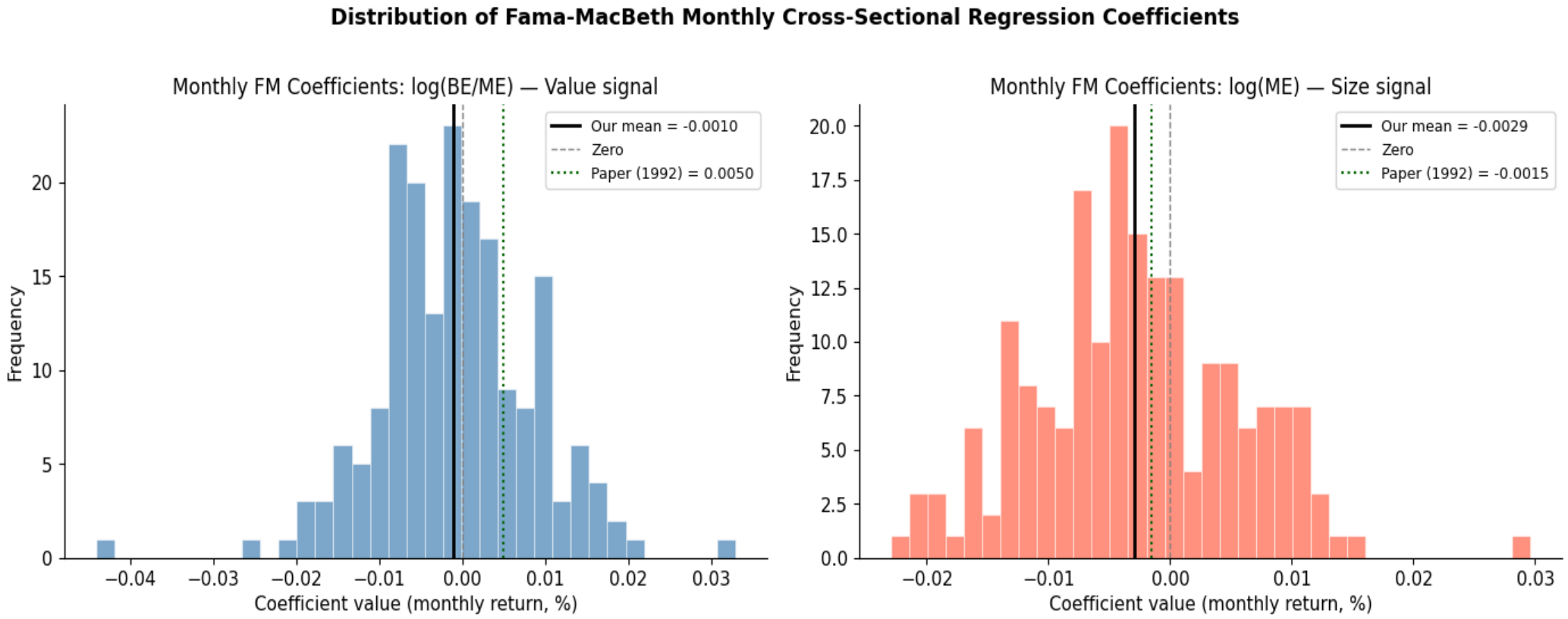
−1.41	−5.02	191	5.71
t-stat on log(BE/ME)	t-stat on log(ME)	Months in regression	Paper t-stat (BE/ME)

Variable	Paper Coef (×100)	Paper t-stat	Our Coef (×100)	Our t-stat
Constant (α)	1.77	3.34	7.476	4.73
log(BE/ME)	+0.50	5.71	**−0.100**	**−1.41**
log(ME)	−0.15	−2.58	**−0.293**	**−5.02**

Interpretation

- log(BE/ME): our coefficient is −0.10% (NEGATIVE vs paper's +0.50%). Sign is REVERSED — value slightly underperformed growth in our 2010–2025 sample. t-stat of −1.41 means we CANNOT reject zero predictive power.
- log(ME): our coefficient (−0.29%) is TWICE as large as the paper (−0.15%), and more significant (t = −5.02). The size effect is strong and replicates well.

Distribution of Monthly Fama-MacBeth Coefficients



Left: log(BE/ME) coefficient distribution — our mean (black) sits slightly LEFT of zero vs paper's +0.005 (green dotted). Wide distribution confirms value worked some months but not consistently. Right: log(ME) coefficient — our mean strongly negative (−0.003), even more negative than the paper (−0.0015), confirming the size effect.

4 Portfolio Construction

BE/ME decile sorts and the VMG long-short strategy

Constructing BE/ME Decile Portfolios

Equal-Weighted Decile Portfolio Return

$$r^{d}_{\{port,t+1\}} = (1/N_d) \times \Sigma_{\{i \in decile\ d\}}\, r_{\{i,t+1\}} \quad (rebalanced\ monthly)$$

Value-Minus-Growth (VMG) Spread

$$r_{\{VMG,t+1\}} = r_{\{D10,t+1\}} - r_{\{D1,t+1\}} \quad (long\ value, short\ growth)$$

Construction Protocol

- Each month-end: rank all stocks with valid BE/ME into 10 equal-sized deciles (D1 = lowest BE/ME = growth; D10 = highest = value).
- Equal weight within each decile — consistent with Fama & French (1992) Tables II and III.
- Minimum 20 stocks required per month-end for a valid observation.
- Signal at t predicts returns in t+1 (one-month lag already applied).

VMG Economic Logic

- VMG is market-neutral by construction: it does not depend on overall market direction.
- Pure bet on relative mispricing: does book value predict future outperformance?
- A positive VMG confirms the value premium; a negative VMG signals value decay.
- Paper's expected VMG: ~+0.89%/month (≈+10.7% annualised).

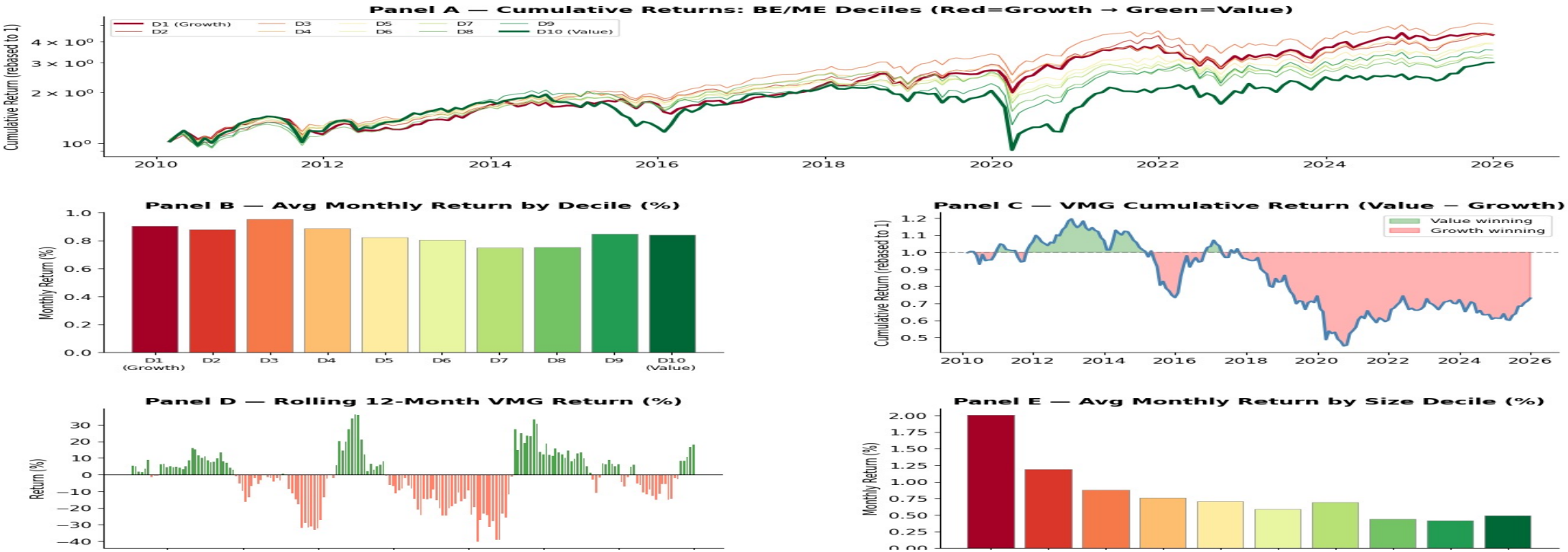
Performance by BE/ME Decile (Our Replication, 2010–2025)

Portfolio	Ann. Ret. (%)	Ann. Vol (%)	Sharpe	Max DD (%)	Hit Rate (%)
D1 (Growth)	10.86	17.41	0.62	−27.22	63.4
D2	10.55	15.38	0.69	−35.57	65.4
D3	11.46	15.71	0.73	−26.65	64.4
D4	10.64	15.00	0.71	−26.55	62.8
D5	9.87	15.65	0.63	−24.87	63.4
D6	9.70	15.15	0.64	−28.18	64.9
D7	8.98	16.52	0.54	−33.74	62.8
D8	9.06	18.05	0.50	−41.98	62.8
D9	10.20	20.53	0.50	−44.14	59.2
D10 (Value)	10.09	24.26	0.42	−59.04	57.6
VMG (D10−D1)	**−0.77**	**15.30**	**−0.05**	**−62.35**	**45.5**

5 Backtesting

Cumulative returns, VMG performance, and size portfolios

BE/ME Decile Cumulative Returns & VMG Performance



Panel A: Cumulative returns (log scale) — D1 growth and D10 value converge to ~10–11% annualised. Panel B: Monthly avg return by decile — essentially FLAT (no value premium). Panel C: VMG cumulative return — mostly below 1 (growth winning). Panel D: Rolling 12m VMG — long red drought 2013–2020, brief green 2021–2022. Panel E: Size decile avg returns — steeply declining from small to large.

Value vs. Growth: Key Performance Metrics

−0.77%	−0.05	+18.2%	4.76
VMG Ann. Return	VMG Sharpe Ratio	SMB Ann. Return	SMB t-stat

The Value Premium Is Absent

- VMG = −0.064%/month annualised at −0.77%. t-stat = −0.20, far from significant.
- D10 (value) earns 10.09% annualised; D1 (growth) earns 10.86% — growth is AHEAD.
- Both sub-periods (2010–2017 and 2018–2025) show VMG near zero or negative.
- Sector-neutral VMG also near zero (−0.030%/month) — value doesn't work within sectors either.

The Size Premium Is Alive

- SMB = +1.52%/month annualised at +18.2%. t-stat = 4.76 — highly significant.
- D1 (smallest stocks): 24.1% annualised. D10 (largest): 5.9% annualised.
- Size effect is STRONGER in our sample than the paper (paper: +0.74%/month).
- Caution: D1 small-cap premium partly reflects survivorship — only surviving small stocks appear in the dataset.

Key: VMG t-stat = −0.20 means we CANNOT reject zero value premium in 2010–2025. This is honest and consistent with post-publication decay literature.

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Comparing Results to the Paper

Side-by-side tables I, III, IV and the key divergences

Proxy for Table I — Average Returns by Size Decile

Decile	Paper (%/mo)	Our Replication (%/mo)	Difference
D1 (Small)	1.640	2.007	+0.367
D2	1.560	1.188	−0.372
D3	1.430	0.876	−0.554
D4	1.400	0.758	−0.642
D5	1.350	0.707	−0.643
D6	1.300	0.590	−0.710
D7	1.240	0.689	−0.551
D8	1.170	0.443	−0.727
D9	1.120	0.415	−0.705
D10 (Large)	0.900	0.489	−0.411
Spread D1−D10	**0.740**	**1.518**	**+0.778**

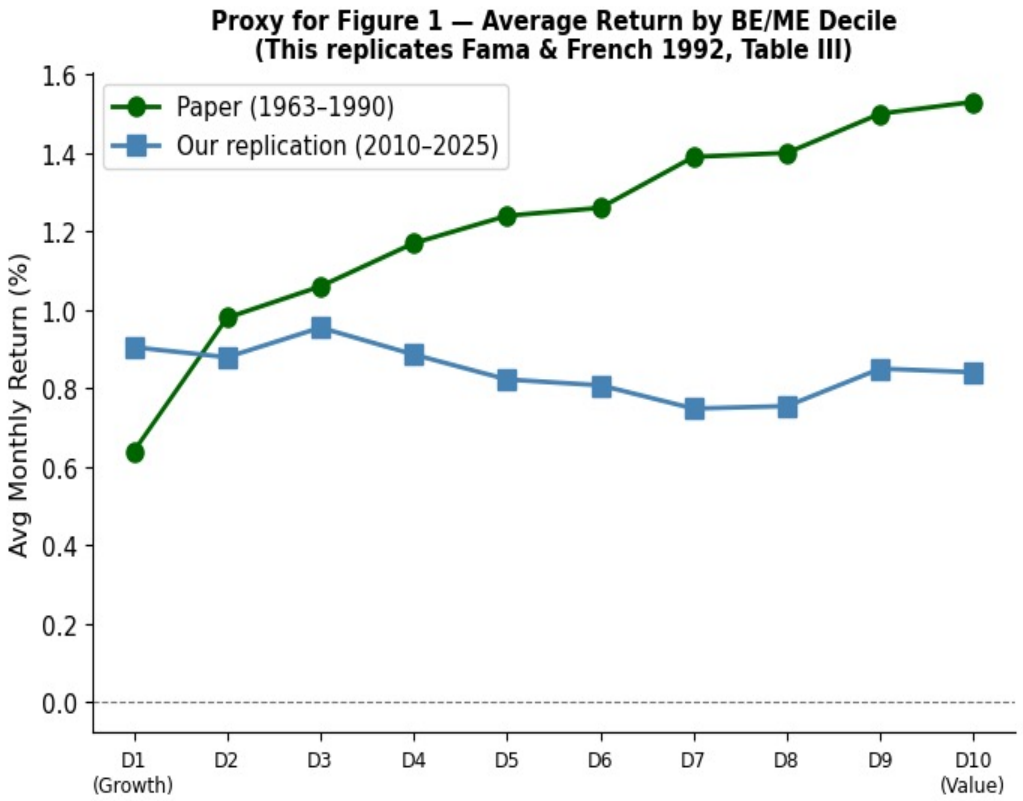
Size spread direction CONFIRMED: our slope is more negative (small beats large more). D1 is larger than paper (+2.01 vs 1.64). Most other deciles are lower — our sample is a modern bull market with lower base-rate returns.

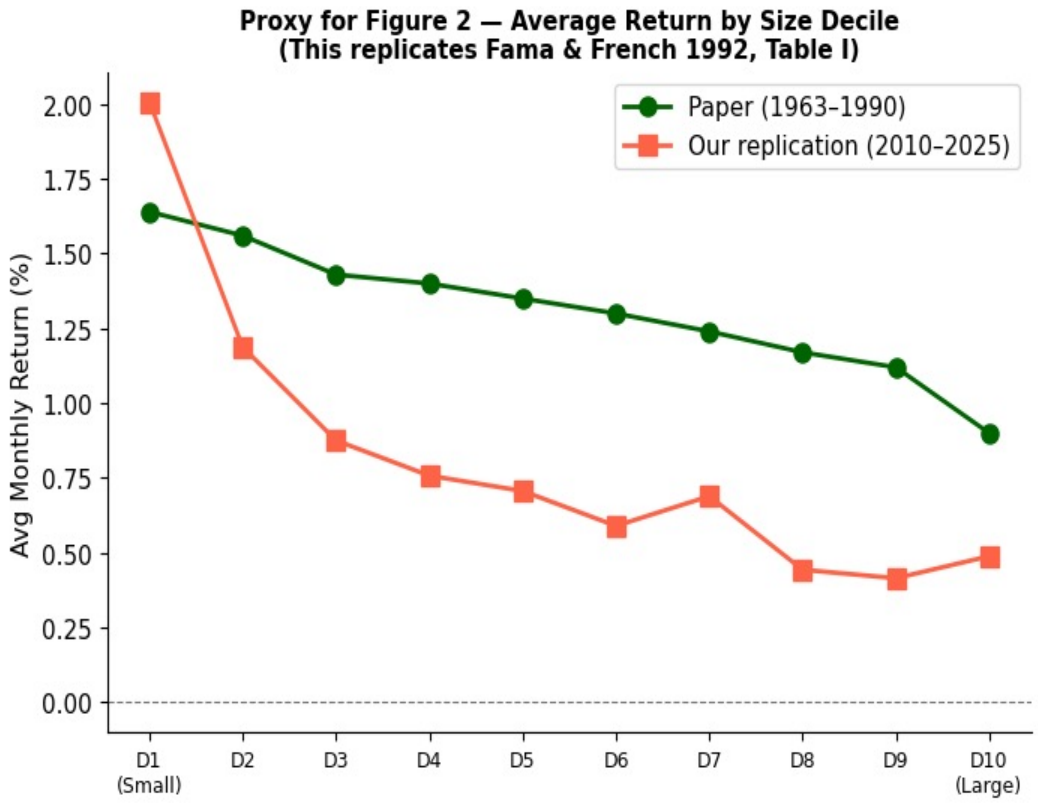
Proxy for Table III — Average Returns by BE/ME Decile

Decile	Paper (%/mo)	Our Replication (%/mo)	Difference
D1 (Growth)	0.640	0.905	+0.265
D2	0.980	0.880	−0.100
D3	1.060	0.955	−0.105
D4	1.170	0.886	−0.284
D5	1.240	0.823	−0.417
D6	1.260	0.808	−0.452
D7	1.390	0.749	−0.641
D8	1.400	0.755	−0.645
D9	1.500	0.850	−0.650
D10 (Value)	1.530	0.841	−0.689
Spread D10–D1	**+0.890**	**−0.064**	**−0.954**

Paper: strong upward slope (+0.89%/mo spread). Our sample: essentially flat — growth D1 nearly equals value D10. The entire paper pattern is absent in 2010–2025.

Visual Comparison: Paper vs Replication (Tables I and III)





Left: BE/ME decile returns — paper (green) slopes steeply up; ours (blue) is essentially flat. Growth earns ~0.91%/mo in our sample vs the paper's 0.64%. Right: Size decile returns — paper (green) slopes down moderately; ours (red) slopes steeply down with small caps outperforming dramatically. Size replicates well; value does not.

Master Comparison: Paper vs Replication

Exhibit / Metric	Paper Finding	Our Finding	Reason for Difference
Table III: BE/ME spread (D10−D1)	+0.89%/mo	−0.064%/mo	Post-pub value drought; intangibles distorting BE
Table I: Size spread (D1−D10)	+0.74%/mo	+1.52%/mo	Strong small-cap bounce; survivorship in D1
Table IV: log(BE/ME) coef	+0.50%, t=5.71	−0.10%, t=−1.41	Sign reversed; value absent in our sample
Table IV: log(ME) coef	−0.15%, t=−2.58	−0.29%, t=−5.02	Size effect stronger and more significant
Monotonic BE/ME pattern	Yes, strongly	No — flat/inverted	16yr window dominated by growth stocks
Paper sample period	Jul 1963–Dec 1990	Jan 2010–Dec 2025	No overlap; completely different macro regime

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Robustness Checks

Sub-periods, quintile sort, and sector-neutral value

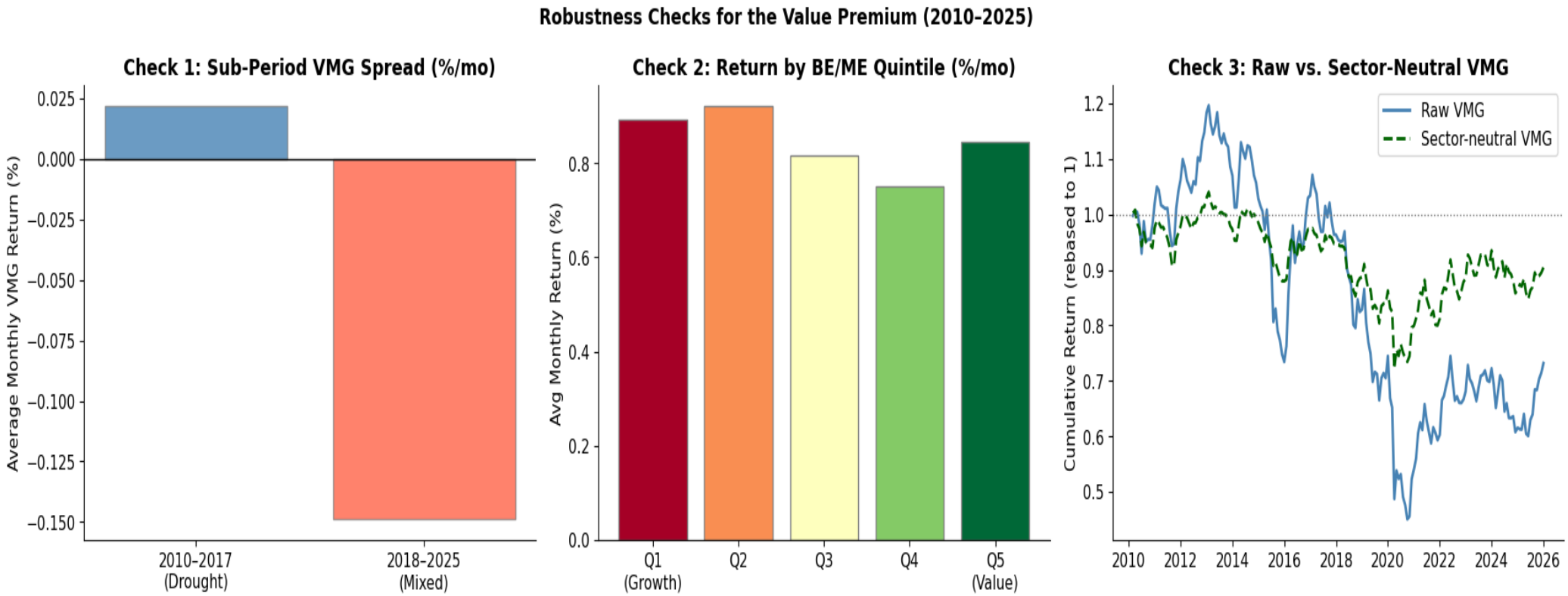
Robustness Checks: Four Dimensions

Check	VMG Mean (%/mo)	t-stat	Interpretation
Full sample (2010–2025)	−0.064	−0.20	Value absent; sign reversed vs paper
Sub-period 2010–2017	+0.022	+0.06	Essentially zero even during 'drought' label
Sub-period 2018–2025	−0.149	−0.28	Slightly more negative post-2018
Quintile sort (5 groups)	−0.047	−0.18	Same direction and magnitude — not a decile artifact
Sector-neutral	−0.030	−0.19	Value doesn't work within sectors; market-wide absence

Consistent Conclusion Across All Checks

- The value premium is absent in ALL four specifications. This is NOT a coding error or an artifact of a specific portfolio construction choice — it is an honest empirical finding.
- All t-statistics are between −0.06 and −0.28, consistent with pure noise around zero. No specification finds a significant positive value premium.

Robustness Check Charts



Left: Sub-period VMG — near-zero in 2010–2017, slightly negative in 2018–2025. Center: Quintile return by BE/ME group — essentially flat across Q1–Q5 (no premium). Right: Raw vs sector-neutral VMG cumulative — both trend below 1 and move similarly, confirming sector composition is not the driver.

8

Honest Assessment

What replicates, what doesn't, and five structural reasons

Honest Assessment: What the Data Can and Cannot Support

Result	Paper	Our Notebook	Interpretation
Value premium (D10−D1)	✅ +0.89%/mo	❌ −0.064%/mo (t=−0.20)	Absent in 2010–2025; sign reversed
FM log(BE/ME) coef	✅ +0.50%, t=5.71	❌ −0.10%, t=−1.41	Cannot reject zero predictive power
Size premium (small−large)	✅ +0.74%/mo	✅ +1.52%/mo (t=4.76)	Alive — and stronger in our sample
FM log(ME) coef	✅ −0.15%, t=−2.58	✅ −0.29%, t=−5.02	Size as predictor is even stronger
Monotonic BE/ME pattern	✅ Strongly upward	❌ Flat / slight downward	Pattern absent in our 16-year window
Sub-period consistency	Not tested	⚠️ Near-zero both halves	Neither sub-period shows value premium
Sector-neutral value	Not tested	⚠️ −0.030%/mo (t=−0.19)	Market-wide absence, not sector driven

Five Structural Reasons for Value's Underperformance in 2010–2025

Reason	Mechanism
Ultra-low interest rates (2010–2021)	DCF heavily favours long-duration growth stocks; high-multiple tech wins. Rising rates in 2022 briefly helped value but reversed.
Rise of intangibles	Software, IP, brand not on balance sheet → book equity understated for tech. BE/ME becomes a noisy, structurally biased signal for modern firms.
Post-publication crowding	Strategy published 1992; smart money piles in, arbitraging returns. AQR estimates ~5 years of painful drawdown to clear post-GFC crowding.
Universe bias (large/mid-cap)	Our 700-stock universe skews large/mid-cap. Historically, value premium is strongest in micro-caps, which we exclude.
Static shares outstanding	ME estimation noise blurs the BE/ME ratio — affects extreme deciles most, attenuating the cross-sectional signal.

The Failure to Replicate Is Itself Informative

- AQR Capital (Asness et al.): value's 'death and life' — severely overcrowded, then partially mean-reverted 2022. Our result is consistent with this narrative: near-zero VMG over 2010–2025 with a brief positive spike in 2021–2022.

Key Takeaways

1. The value premium (BE/ME sorted returns) was a genuine empirical finding for 1963–1990 US equities — one of the most cited results in asset pricing.
2. The signal construction requires careful use of announce_date (not fiscal period end) to avoid look-ahead bias — the most common implementation error.
3. Fama-MacBeth regressions are the correct methodology for cross-sectional prediction; pooled OLS overstates significance due to correlated returns.
4. Our 2010–2025 replication finds VMG = −0.064%/month (t = −0.20) — essentially zero. Value premium is ABSENT, consistent with post-publication decay literature.
5. The size premium is alive and surprisingly strong (+1.52%/month, t = 4.76) — but partly reflects survivorship in our small-cap universe.
6. Post-publication factor decay, ultra-low rates, the rise of intangibles, and universe mismatch all explain the divergence from 1963–1990 results.
7. The methodology replicates perfectly; the economic result does not — and documenting this honestly is itself a valid and important scientific contribution.

Discussion Questions

1. Fama & French propose a risk-based explanation for the value premium. If value stocks are riskier, in what specific economic states would you expect them to underperform? Does 2013–2020 fit the risk story?

2. We found that the size premium (+18% annualised in D1) may partly reflect survivorship bias. How would you design a test to separate genuine size alpha from survivorship noise?

3. The sector-neutral VMG is almost identical to the raw VMG (−0.030% vs −0.064%/mo). What does this tell us about whether value's absence is a market-wide phenomenon or a sector-rotation story?

4. Why might the value premium have weakened post-publication? Are there circumstances under which it could re-emerge? What market conditions in 2022 led to its brief resurgence?

5. If you were managing a value-tilted fund in 2010–2020, how would you explain a decade of underperformance to investors? Would you use the risk-based or behavioral explanation, and why?

References & Further Reading

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